

Supplementary Materials for Scalable Nonparametric Bayesian Method for Clinical Risk Prediction Using Large-Scale Data from Heterogeneous Populations

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A Subset posterior inference of designed JDP-LNMM model

JDP has two layers of DP prior. The first layer, called the θ -layer, controls the θ -clusters and posterior of $y_i|\mathbf{x}_i, \boldsymbol{\theta}_i$; the second layer, called the ψ -layer within θ -layer, controls the ψ -subclusters within each θ -cluster and the posterior of $\mathbf{x}_i|\boldsymbol{\psi}_i$. These two layers allows the usage of more kernels across the θ -clusters to describe the marginal of $\mathbf{x}$ and furthermore, the conditional of $y|\mathbf{x}$. α_θ and α_ψ are the concentration parameters of these two layers, respectively.

Now, we design the posterior computation algorithm. Suppose that this subset has n observations. Let $\mathbf{s}_i = (s_{y,i}, s_{x,i})$ be the predictive clustering allocation of the i th observation pair in θ -clusters and ψ -subclusters, and $\boldsymbol{\rho} = \{(s_{y,i}, s_{x,i})\}_{i=1}^n$ be the recorded clustering allocation, $\tilde{n}$ be the number of θ -clusters, and $\mathcal{D}_j = (n_j, \mathcal{X}_j, \mathcal{Y}_j)$ be the information of θ -cluster j including the number of observation pairs, the response and the covariates in this θ -cluster. Usually, the ψ -concentration parameter α_ψ is set to be independent of α_θ . Therefore, for the ψ -layer, let $\mathcal{D}_{x,j} = (n_{x,j}, \mathcal{X}_j)$ denote the information of ψ -cluster j including the number of observations and the covariates in this ψ -cluster. For the conditional and marginal parameters, let $\boldsymbol{\Theta} = \{\boldsymbol{\theta}_i\}_{i=1}^n$ be the set of parameters for all the response conditionals and $\boldsymbol{\Psi} = \{\boldsymbol{\psi}_i\}_{i=1}^n$ be the set of parameters for all the covariate marginals. Moreover, two superscripts are introduced, $-i$ is introduced to all these notations representing the similar meaning when the i th pair $(y_i, \mathbf{x}_i)$ is excluded. Let parameters with superscript $*$ denote those associated with the clusters. In the following two subsections, algorithms from both marginal and conditional methods are provided.

A.1 Marginal Method

Algorithm 1 gives an overview of each iteration in the Gibbs sampling algorithm of three main steps in the marginal method. Hyper-parameters are omitted in algorithm 1. The first step is to update the cluster labels, which also determines if a new cluster is needed. The second step is to update cluster posterior sample ($\boldsymbol{\Theta}$ and $\boldsymbol{\Psi}$) based on current cluster allocation $\boldsymbol{\rho}$. The final step is to update the concentration parameters (α_θ and α_ψ) and the hyper-parameters.

Algorithm 1 Marginal Method: Gibbs Sampling Algorithm for JDP-LNMM

Input:

Dataset $\mathcal{D} = (\mathcal{X}, \mathcal{Y})$.

Output:

Cluster structure $\rho = \{s_i : i = 1, 2, \dots, n\}$;

θ -layer posteriors $\Theta = \{(\beta_j, \sigma_j^2)\}_{j=1}^{\max_i s_{y,i}}$;

ψ -layer posteriors $\Psi = \{(\mu_{x,j}, \Sigma_{x,j})\}_{j=1}^{\max_i s_{x,i}}$.

Initialize: ρ, Θ, Ψ

- 1: **Step 1:**
 - 2: **for** $i = 1$ **to** n **do:**
 - 3: **Sample** s_i **from** $f(s_i | \rho^{-i}, \Theta^{-i}, \Psi^{-i}, \mathcal{D}^{-i})$;
 - 4: **Step 2:**
 - 5: **for** $j = 1$ **to** $\max_i s_{y,i}$ **do:**
 - 6: **Sample** β_j **from** $f(\beta_j | \sigma_j^2, \rho, \mathcal{D}_j)$;
 - 7: **Sample** σ_j^2 **from** $f(\sigma_j^2 | \beta_j, \rho, \mathcal{D}_j)$;
 - 8: **for** $j = 1$ **to** $\max_i s_{x,i}$ **do:**
 - 9: **Sample** $\mu_{x,j}$ **from** $f(\mu_{x,j} | \Sigma_{x,j}, \rho, \mathcal{D}_{x,j})$;
 - 10: **Sample** $\Sigma_{x,j}$ **from** $f(\Sigma_{x,j} | \mu_{x,j}, \rho, \mathcal{D}_{x,j})$;
 - 11: **Step 3:**
 - 12: **Sample** $(\alpha_\psi, \alpha_\theta)$ **from** $f(\alpha_\psi, \alpha_\theta | \rho)$;
 - 13: **Update** hyper-parameters.
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The computational details in the Algorithm 1 are provided as follows:

Line 3: s_i from $f(s_i | \rho^{-i}, \Theta^{-i}, \Psi^{-i}, \mathcal{D}^{-i})$.

We first sample $s_{y,i}$ from

$$s_{y,i} \propto \begin{cases} \alpha_\theta \int k(y_i | \mathbf{x}_i, \boldsymbol{\theta}) H_\theta(d\boldsymbol{\theta}) \int k(\mathbf{x}_i | \boldsymbol{\psi}) H_\psi(d\boldsymbol{\psi}), & s_{y,i} = \max_i s_{y,i} + 1; \\ n_j^{-i} [\alpha_\psi k(y_i | \mathbf{x}_i, \boldsymbol{\theta}_j^*) \int k(\mathbf{x}_i | \boldsymbol{\psi}) H_\psi(d\boldsymbol{\psi}) + \\ \sum_l n_{jl}^{-i} k(y_i | \mathbf{x}_i, \boldsymbol{\theta}_j^*) k(\mathbf{x}_i | \boldsymbol{\psi}_l^*)], & s_{y,i} = j, \end{cases}$$

where n_{jl}^{-i} denotes the number of observations in θ -cluster j that are also assigned to ψ -cluster l , excluding the i th observation. Then, $s_{x,i}$ is sampled from

$$s_{x,i} \propto \begin{cases} \alpha_\psi \int k(\mathbf{x}_i | \boldsymbol{\psi}) H_\psi(d\boldsymbol{\psi}), & s_{x,i} = \max_i s_{x,i} + 1; \\ n_{x,l}^{-i} k(\mathbf{x}_i | \boldsymbol{\psi}_l^*), & s_{x,i} = l. \end{cases}$$

It is remarkable that above equations are for noncensored data. For censored data, all the $k(y_i | \mathbf{x}_i, \boldsymbol{\theta})$ will be replaced by $1 - K(y_i | \mathbf{x}_i, \boldsymbol{\theta})$. The analytical form of these integrals are given as:

$$\int k(\mathbf{x}_i | \boldsymbol{\psi}) H_\psi(d\boldsymbol{\psi}) = \prod_{h=1}^p \frac{1}{\sqrt{2\pi}} \cdot \frac{c_h}{1 + c_h} \cdot \frac{\Gamma(A_h)}{\Gamma(a_{x,h})} \cdot b_{x,h}^{a_{x,h}} \cdot B_h^{-A_h},$$

where $A_h = a_{x,h} + \frac{1}{2}$ and $B_h = b_{x,h} + \frac{c_h}{2(1+c_h)} (x_h - \mu_{0\psi,h})^2$, and

$$\int k(\mathbf{y}_i | \mathbf{x}_i, \boldsymbol{\theta}) H_{\boldsymbol{\theta}}(d\boldsymbol{\theta}) = \sqrt{\frac{\det(\mathbf{C})}{2\pi \det(\mathbf{x}\mathbf{x}^\top + \mathbf{C})}} \cdot \frac{\Gamma(A_\sigma)}{\Gamma(a_\sigma)} \cdot b_\sigma^{a_\sigma} \cdot B_\sigma^{-A_\sigma},$$

where $A_\sigma = a_\sigma + \frac{1}{2}$ and $B_\sigma = b_\sigma - \frac{1}{2} (\mathbf{y}\mathbf{x}^\top + \boldsymbol{\mu}_{0\theta}^\top \mathbf{C}) (\mathbf{x}\mathbf{x}^\top + \mathbf{C})^{-1} (\mathbf{y}\mathbf{x}^\top + \boldsymbol{\mu}_{0\theta}^\top \mathbf{C})^\top$.

Line 6 ℰ 7: (β_j, σ_j^2) from $f(\beta_j | \sigma_j^2, \boldsymbol{\rho}, \mathcal{D}_j) f(\sigma_j^2 | \beta_j, \boldsymbol{\rho}, \mathcal{D}_j)$.

The conjugate posterior of β_j is

$$\mathcal{N}(\boldsymbol{\mu}_n, \sigma_j^2 \mathbf{C}_n^{-1}),$$

where $\mathbf{C}_n = \mathcal{X}_j^\top \mathcal{X}_j + \mathbf{C}$, and $\boldsymbol{\mu}_n = \mathbf{C}_n^{-1} (\mathcal{X}_j^\top \mathcal{X}_j \beta_j + \mathbf{C} \boldsymbol{\mu}_{0\theta})$.

The conjugate posterior of σ_j^2 is

$$\text{IG}(a_{n,\theta}, b_{n,\theta}),$$

where $a_{n,\theta} = a_\theta + \frac{n_j}{2}$, and $b_{n,\theta} = b_\theta + \frac{1}{2} (\mathbf{y}_j^\top \mathbf{y}_j + \boldsymbol{\mu}_{0\theta}^\top \mathbf{C} \boldsymbol{\mu}_{0\theta} - \boldsymbol{\mu}_n^\top \mathbf{C}_n \boldsymbol{\mu}_n)$.

Line 9 ℰ 10: $(\boldsymbol{\mu}_{x,j}, \boldsymbol{\Sigma}_{x,j})$ from $f(\boldsymbol{\mu}_{x,j} | \boldsymbol{\Sigma}_{x,j}, \boldsymbol{\rho}, \mathcal{X}_j) f(\boldsymbol{\Sigma}_{x,j} | \boldsymbol{\rho}, \mathcal{X}_j)$.

Notice that $\boldsymbol{\mu}_{x,j} = \{\mu_{x,h,j}\}_{h=1}^p$ and $\boldsymbol{\Sigma}_{x,j} = \text{diag}\{\sigma_{x,1,j}^2 c_{x,1,j}^{-1}, \dots, \sigma_{x,p,j}^2 c_{x,p,j}^{-1}\}$.

For $h = 1, \dots, p$, omitting the subscripts, the posterior of $(\mu_{x,h,j}, \sigma_{x,h,j}^2)$ is

$$\begin{aligned} p(\mu, \sigma^2 | \boldsymbol{\rho}, \mathcal{D}_{x,j}) &\propto p(\mathcal{X}_{j,h} | \mu, \sigma^2) p(\mu | \sigma^2) p(\sigma^2) \\ &\propto p(\mu | \sigma^2, \mathcal{X}_{j,h}) p(\sigma^2 | \mathcal{X}_{j,h}). \\ &\propto \mathcal{N}(\mu_{n,h}, \sigma_{n,h}^2) \times \text{IG}(a_{n,h}, b_{n,h}), \end{aligned}$$

where $\sigma_{n,h}^2 = \frac{\sigma^2}{n_j + c_h}$, $a_{n,h} = a_{x,h} + \frac{n_j}{2}$, $\mu_{n,h} = \frac{c_h \mu_{0\psi,h} + \sum_{i=1}^{n_j} x_i}{c_h + n_j}$, and $b_{n,h} = b_{x,h} + \frac{1}{2} \sum_{i=1}^{n_j} (x_i - \mu_{n,h})^2$.

Remark: The α_θ , α_ψ , and hyper-parameters are updated from their conjugate posteriors.

B Details on the simulated data

In this section, we provide the parameters we used to generate the simulated data in the first two scenarios. In each component of each scenario, we first draw $\mathbf{e}$ from the following independent MVN distribution:

$$\mathbf{e} = [e_1, e_2, \dots, e_{50}]^\top \sim MVN(\mathbf{1}_{50}, \sigma_x^2 \mathbf{I}).$$

Then, the covariate $\mathbf{x}$ is obtained by scaling $\mathbf{e}$ using the mean vector $\boldsymbol{\mu}_x$ in each dimension as

$$\mathbf{x} = [x_1, x_2, \dots, x_{50}]^\top = [\mu_{x1}e_1, \mu_{x2}e_2, \dots, \mu_{x50}e_{50}]^\top.$$

B.1 Scenario 1: $J = 3$

The mean vector $\boldsymbol{\mu}_{x,1}$ for covariates in component 1 is

$$\boldsymbol{\mu}_{x,1} = \begin{bmatrix} 7.29 & -1.94 & 7.58 & 0.45 & 2.12 & 3.82 & 6.75 & 5.1 & 6.34 & -1.91 \\ 5.8 & 8.68 & 5.43 & -1.5 & 1.05 & 8. & 4.06 & -0.48 & 4.75 & 5.04 \\ 1.76 & -1.14 & 8.41 & 0.74 & 4.28 & -1.28 & -2.11 & 8.34 & 6.51 & 3.6 \\ 0.92 & 1.61 & -1.55 & 7.51 & 7.9 & -2.62 & 4.79 & -2.77 & -0.18 & 3.43 \\ 3.06 & 1.01 & -2.08 & 0.43 & 7.05 & 5.33 & 8.26 & 3.22 & 8.1 & 4.74 \end{bmatrix}^\top,$$

and the variance $\sigma_{x,1}^2$ is 0.09.

The coefficient vector $\boldsymbol{\beta}_1$ in component 1 is

$$\boldsymbol{\beta}_1 = \begin{bmatrix} 0.053 & -0.014 & 0.055 & 0.003 & 0.015 & 0.028 & 0.049 & 0.037 & 0.046 & -0.014 \\ 0.042 & 0.063 & 0.04 & -0.011 & 0.008 & 0.058 & 0.03 & -0.004 & 0.035 & 0.037 \\ 0.013 & -0.008 & 0.061 & 0.005 & 0.031 & -0.009 & -0.015 & 0.061 & 0.048 & 0.026 \\ 0.007 & 0.012 & -0.011 & 0.055 & 0.058 & -0.019 & 0.035 & -0.02 & -0.001 & 0.025 \\ 0.022 & 0.007 & -0.015 & 0.003 & 0.052 & 0.039 & 0.06 & 0.024 & 0.059 & 0.035 \end{bmatrix}^\top,$$

and the variance σ_1^2 is 0.11.

The mean vector $\boldsymbol{\mu}_{x,2}$ for covariates in component 2 is

$$\boldsymbol{\mu}_{x,2} = \begin{bmatrix} 0.22 & -5.3 & -2.8 & -2.65 & -0.03 & -6.31 & -1.95 & 0.22 & -0.02 & -5.16 \\ -4.65 & -6.66 & -7.78 & 0.07 & -7.94 & 1.1 & -9.2 & -3.97 & -7.37 & 0.03 \\ -3.99 & -3.29 & -6.31 & -0.49 & -3.85 & -8.41 & -6.54 & -9.04 & -4.04 & -4.44 \\ 0.23 & 0.66 & -1.94 & -0.4 & 0.49 & -0.69 & -0.41 & -0.99 & -7.7 & -6.41 \\ -6.25 & -7.95 & -3.34 & -0.21 & -6.05 & -6.57 & -2.17 & -7.48 & -3.26 & -0.22 \end{bmatrix}^\top,$$

and the variance $\sigma_{x,2}^2$ is 0.17.

The coefficient vector $\boldsymbol{\beta}_2$ in component 2 is

$$\boldsymbol{\beta}_2 = \begin{bmatrix} 0.001 & -0.027 & -0.014 & -0.014 & -0.0 & -0.033 & -0.01 & 0.001 & -0.0 & -0.027 \\ -0.024 & -0.034 & -0.04 & 0.0 & -0.041 & 0.006 & -0.047 & -0.021 & -0.038 & 0.0 \\ -0.021 & -0.017 & -0.033 & -0.003 & -0.02 & -0.043 & -0.034 & -0.047 & -0.021 & -0.023 \\ 0.001 & 0.003 & -0.01 & -0.002 & 0.003 & -0.004 & -0.002 & -0.005 & -0.04 & -0.033 \\ -0.032 & -0.041 & -0.017 & -0.001 & -0.031 & -0.034 & -0.011 & -0.039 & -0.017 & -0.001 \end{bmatrix}^\top,$$

and the variance σ_2^2 is 0.11.

The mean vector $\boldsymbol{\mu}_{x,3}$ for covariates in component 3 is

$$\boldsymbol{\mu}_{x,3} = \begin{bmatrix} 5.31 & 4.59 & 2.25 & 0.72 & 1.14 & 0.37 & 0.75 & 1.18 & 4.51 & -0.02 \\ -0.12 & 3.84 & 3.7 & 1.26 & 1.16 & 3.26 & 0.37 & 2.58 & -0.23 & 1.03 \\ 5.1 & -0.14 & 2.63 & 5.24 & 2.21 & 3.26 & -0.81 & 1.9 & -0.03 & 2.03 \\ 0.06 & 4.73 & 1.37 & 1.41 & 4.49 & 3.12 & 2.72 & 3.27 & 0.12 & 4.43 \\ 1.82 & 4.32 & 3.88 & 4.88 & -0.57 & -0.02 & 3.07 & 2.54 & 0.21 & 1.71 \end{bmatrix}^\top,$$

and the variance $\sigma_{x,3}^2$ is 0.25.

The coefficient vector $\boldsymbol{\beta}_3$ in component 3 is

$$\boldsymbol{\beta}_3 = \begin{bmatrix} 0.103 & 0.089 & 0.044 & 0.014 & 0.022 & 0.007 & 0.015 & 0.023 & 0.088 & 0.0 \\ -0.002 & 0.075 & 0.072 & 0.024 & 0.023 & 0.063 & 0.007 & 0.05 & -0.004 & 0.02 \\ 0.099 & -0.003 & 0.051 & 0.102 & 0.043 & 0.063 & -0.016 & 0.037 & -0.001 & 0.039 \\ 0.001 & 0.092 & 0.027 & 0.027 & 0.087 & 0.061 & 0.053 & 0.063 & 0.002 & 0.086 \\ 0.035 & 0.084 & 0.075 & 0.095 & -0.011 & -0.001 & 0.06 & 0.049 & 0.004 & 0.033 \end{bmatrix}^\top,$$

and the variance σ_3^2 is 0.11.

B.2 Scenario 2: $J = 5$

The mean vector $\boldsymbol{\mu}_{x,1}$ for covariates in component 1 is

$$\boldsymbol{\mu}_{x,1} = \begin{bmatrix} -5.03 & 1.04 & 5.16 & 2.08 & 2.8 & 3.53 & 4.8 & -4.08 & 4.62 & 1.06 \\ 4.39 & -5.63 & 4.23 & -1.23 & 2.34 & 5.34 & 3.63 & 1.67 & 3.94 & 4.06 \\ -2.64 & -1.39 & 5.52 & 2.2 & 3.73 & 1.33 & 0.97 & 5.48 & -4.69 & 3.44 \\ 2.28 & 2.58 & 1.21 & 5.13 & 5.29 & 0.75 & -3.95 & 0.69 & 1.8 & 3.36 \\ 3.2 & 2.32 & 0.98 & 2.07 & -4.93 & 4.18 & 5.45 & 3.27 & -5.38 & 3.93 \end{bmatrix}^\top,$$

and the variance $\sigma_{x,1}^2$ is 0.05.

The coefficient vector $\boldsymbol{\beta}_1$ in component 1 is

$$\boldsymbol{\beta}_1 = \begin{bmatrix} -0.065 & 0.014 & 0.067 & 0.027 & 0.036 & 0.046 & 0.062 & -0.053 & 0.06 & 0.014 \\ 0.057 & -0.073 & 0.055 & -0.016 & 0.03 & 0.069 & 0.047 & 0.022 & 0.051 & 0.053 \\ -0.034 & -0.018 & 0.072 & 0.029 & 0.048 & 0.017 & 0.013 & 0.071 & -0.061 & 0.045 \\ 0.03 & 0.033 & 0.016 & 0.067 & 0.069 & 0.01 & -0.051 & 0.009 & 0.023 & 0.044 \\ 0.042 & 0.03 & 0.013 & 0.027 & -0.064 & 0.054 & 0.071 & 0.042 & -0.07 & 0.051 \end{bmatrix}^\top,$$

and the variance σ_1^2 is 0.11.

The mean vector $\boldsymbol{\mu}_{x,2}$ for covariates in component 2 is

$$\boldsymbol{\mu}_{x,2} = \begin{bmatrix} -9.92 & 6.25 & 8.36 & 8.49 & 9.71 & -5.4 & -9.08 & -9.91 & 9.71 & 6.37 \\ 6.8 & -5.1 & -4.15 & 9.79 & -4.02 & -9.66 & 2.96 & -7.37 & 4.5 & 9.76 \\ 7.36 & -7.95 & 5.4 & 9.31 & 7.47 & -3.62 & 5.2 & 3.09 & 7.31 & 6.98 \\ -9.93 & 9.28 & -9.09 & 9.39 & 9.14 & 9.15 & 9.38 & 9.89 & 4.23 & 5.31 \\ -5.44 & 4.01 & 7.91 & 9.55 & 5.62 & -5.18 & 8.9 & 4.41 & 7.97 & 9.54 \end{bmatrix}^\top,$$

and the variance $\sigma_{x,2}^2$ is 0.1.

The coefficient vector β_2 in component 2 is

$$\beta_2 = \begin{bmatrix} -0.025 & 0.014 & 0.019 & 0.019 & 0.024 & -0.012 & -0.021 & -0.025 & 0.024 & 0.015 \\ 0.016 & -0.012 & -0.009 & 0.025 & -0.009 & -0.027 & 0.007 & -0.017 & 0.01 & 0.025 \\ 0.017 & -0.018 & 0.012 & 0.024 & 0.017 & -0.008 & 0.012 & 0.007 & 0.017 & 0.016 \\ -0.025 & 0.026 & -0.021 & 0.024 & 0.025 & 0.023 & 0.024 & 0.023 & 0.01 & 0.012 \\ -0.012 & 0.009 & 0.018 & 0.024 & 0.013 & -0.012 & 0.02 & 0.01 & 0.018 & 0.024 \end{bmatrix}^\top,$$

and the variance σ_2^2 is 0.11.

The mean vector $\mu_{x,3}$ for covariates in component 3 is

$$\mu_{x,3} = \begin{bmatrix} 5.56 & 4.94 & -2.95 & 1.64 & 2.0 & 1.34 & 1.67 & -2.03 & 4.88 & 1.01 \\ 0.92 & 4.3 & -4.19 & 2.1 & -2.02 & 3.81 & -1.34 & -3.23 & 0.83 & 1.9 \\ -5.38 & -0.9 & -3.27 & 5.5 & -2.91 & 3.81 & 0.34 & 2.65 & 1.0 & 2.76 \\ 1.08 & 5.07 & -2.2 & 2.23 & 4.86 & -3.69 & -3.35 & -3.82 & 1.12 & 4.81 \\ 2.58 & 4.71 & 4.34 & -5.2 & 0.54 & 1.0 & 3.64 & 3.2 & 1.2 & 2.48 \end{bmatrix}^\top,$$

and the variance $\sigma_{x,3}^2$ is 0.15.

The coefficient vector β_3 in component 3 is

$$\beta_3 = \begin{bmatrix} 0.038 & 0.034 & -0.02 & 0.011 & 0.014 & 0.009 & 0.011 & -0.014 & 0.034 & 0.007 \\ 0.006 & 0.03 & -0.029 & 0.014 & -0.014 & 0.026 & -0.009 & -0.022 & 0.006 & 0.013 \\ -0.037 & -0.006 & -0.023 & 0.038 & -0.02 & 0.026 & 0.002 & 0.018 & 0.007 & 0.019 \\ 0.007 & 0.035 & -0.015 & 0.015 & 0.034 & -0.025 & -0.023 & -0.026 & 0.008 & 0.033 \\ 0.018 & 0.033 & 0.03 & -0.036 & 0.004 & 0.007 & 0.025 & 0.022 & 0.008 & 0.017 \end{bmatrix}^\top,$$

and the variance σ_3^2 is 0.11.

The mean vector $\mu_{x,4}$ for covariates in component 4 is

$$\mu_{x,4} = \begin{bmatrix} 4.55 & 5.32 & -1.75 & 2.79 & -6.9 & 1.69 & 1.99 & 5.43 & 4.89 & 5.62 \\ 1.5 & 1.27 & -6.69 & 4.62 & -1.44 & 4.29 & -4.62 & -6.7 & -4.36 & 3.14 \\ -3.63 & 2.13 & 4.33 & 2.09 & 4.42 & 3.78 & -4.02 & 5.86 & 5.31 & 1.33 \\ 4.95 & 4.78 & -5.89 & -2.14 & -4.62 & 4.62 & -2.72 & 6.15 & 4.46 & 6.81 \\ 3.57 & 1.68 & 3.97 & 3.51 & 6.41 & 5.17 & -6.26 & 4.45 & 4.74 & 2.97 \end{bmatrix}^\top,$$

and the variance $\sigma_{x,4}^2$ is 0.2.

The coefficient vector β_4 in component 4 is

$$\beta_4 = \begin{bmatrix} 0.027 & 0.032 & -0.011 & 0.017 & -0.041 & 0.01 & 0.012 & 0.033 & 0.029 & 0.034 \\ 0.009 & 0.008 & -0.04 & 0.028 & -0.009 & 0.026 & -0.028 & -0.04 & -0.026 & 0.019 \\ -0.022 & 0.013 & 0.026 & 0.013 & 0.027 & 0.023 & -0.024 & 0.035 & 0.032 & 0.008 \\ 0.03 & 0.029 & -0.035 & -0.013 & -0.028 & 0.028 & -0.016 & 0.037 & 0.027 & 0.041 \\ 0.021 & 0.01 & 0.024 & 0.021 & 0.039 & 0.031 & -0.038 & 0.027 & 0.028 & 0.018 \end{bmatrix}^\top,$$

and the variance σ_4^2 is 0.11.

The mean vector $\mu_{x,5}$ for covariates in component 5 is

$$\mu_{x,5} = \begin{bmatrix} -4.37 & 9.3 & 5.41 & -4.78 & -9.18 & 9.18 & 9.32 & -9.8 & 9.31 & 9.54 \\ 9.65 & 9.61 & -9.67 & 9.92 & -7.98 & 9.26 & 7.69 & -9.52 & 9.13 & 5.24 \\ 9.77 & -7.22 & 7.28 & -9.86 & 4.46 & -9.55 & 9.99 & 8.19 & -6.41 & -9.06 \\ -9.96 & -7.79 & -9.11 & 4.65 & 9.28 & -7.88 & 9.81 & 9.49 & 8.03 & 9.0 \\ 5.09 & 4.7 & -7.26 & 7.94 & 9.21 & -9.01 & -9.06 & 9.04 & 5.08 & 9.33 \end{bmatrix}^\top,$$

and the variance $\sigma_{x,5}^2$ is 0.25.

The coefficient vector β_5 in component 5 is

$$\beta_5 = \begin{bmatrix} -0.007 & 0.015 & 0.009 & -0.008 & -0.018 & 0.015 & 0.022 & -0.018 & 0.022 & 0.019 \\ 0.021 & 0.021 & -0.021 & 0.023 & -0.013 & 0.015 & 0.013 & -0.017 & 0.023 & 0.009 \\ 0.019 & -0.012 & 0.012 & -0.019 & 0.007 & -0.016 & 0.021 & 0.013 & -0.01 & -0.015 \\ -0.016 & -0.013 & -0.023 & 0.008 & 0.02 & -0.013 & 0.019 & 0.022 & 0.013 & 0.023 \\ 0.008 & 0.008 & -0.012 & 0.013 & 0.015 & -0.018 & -0.018 & 0.016 & 0.008 & 0.022 \end{bmatrix}^\top,$$

and the variance σ_y^2 is 0.11.

C Sensitivity analysis on the number of components

In this section, the details of a sensitivity analysis on the synthetic data from an increasing number of components are shown. All of the following results are based on 5-fold experiments. Figure S1 shows the density of the simulated data in all scenarios.

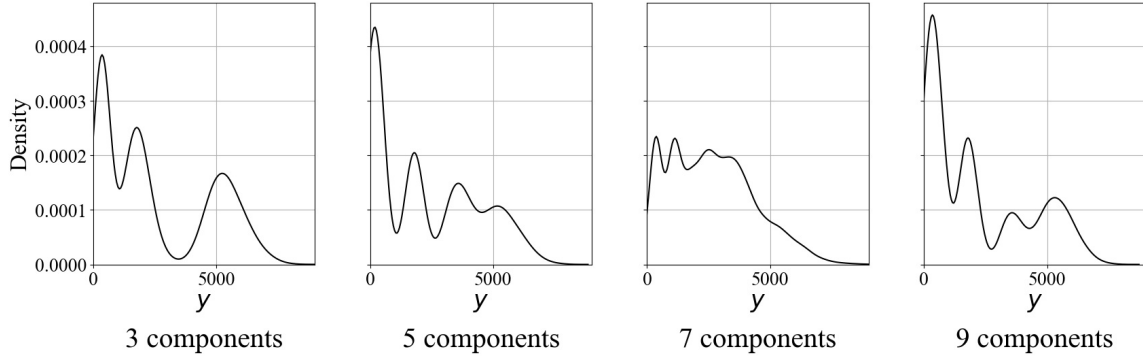
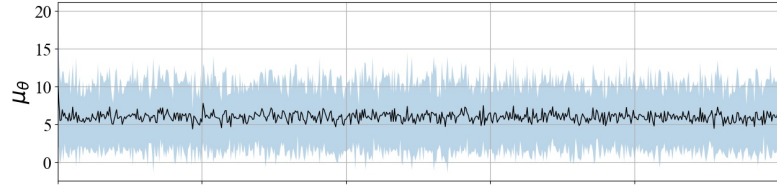


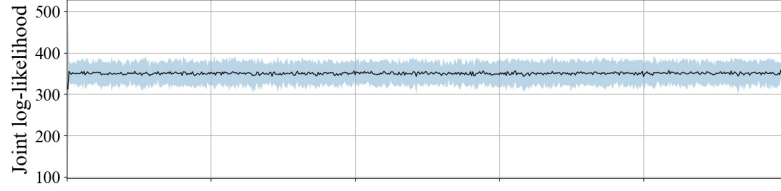
Figure S1: Density of simulated data.

For each scenario, we first show convergence check before posterior collection, the Zeppelin plot of the true and predicted survival probabilities and the density plots for selected θ 's. Then, we summarized the aggregated clustering and predictive performance compared to the benchmark models. Finally, the subset performance and subset Zeppelin plots are given.

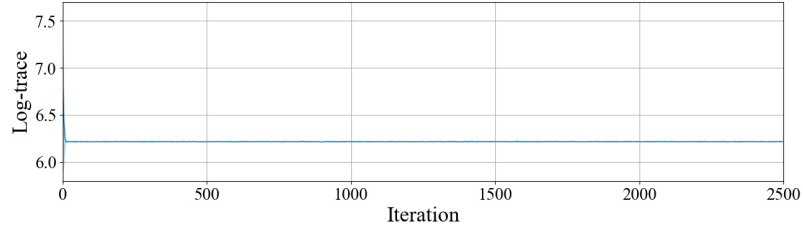
C.1 Scenario 1: $J = 3$



(a) Trend of μ_θ .

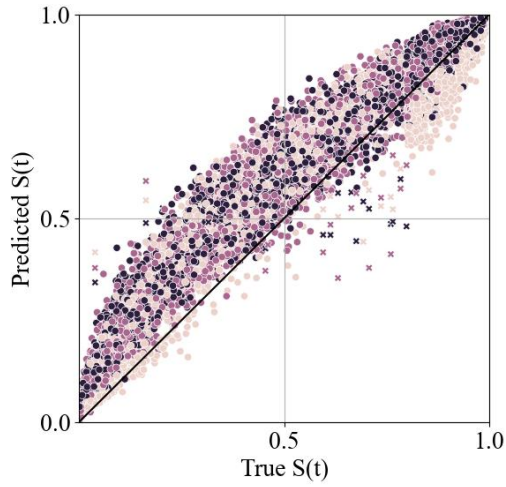


(b) Trend of joint log-likelihood.

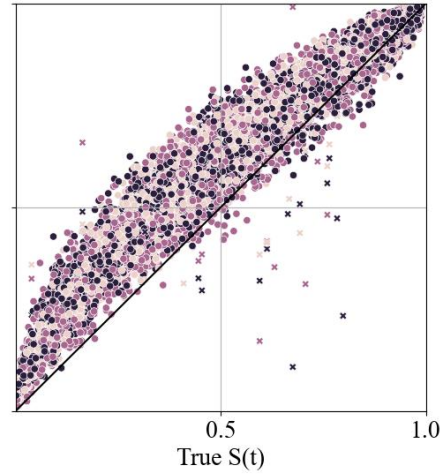


(c) Trend of log-trace of clustering allocations.

Figure S2: Convergence check.



(a) From unit-level aggregator



(b) From cluster-level aggregator

Figure S3: Predictions V.s. True survival probabilities.

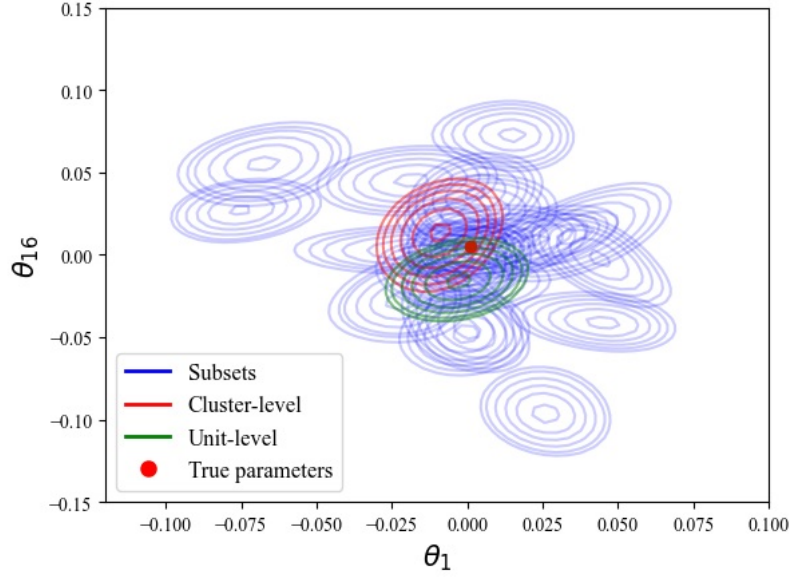


Figure S4: Kernel density estimates of subset posterior distributions and aggregated posterior distributions from cluster- and unit-level aggregators for coefficients (θ_1, θ_{16}) of cluster 2.

Table S4: Clustering Results from the proposed model and benchmarks.

		AR		AM		VM		FM	
		Train	Test	Train	Test	Train	Test	Train	Test
KM	$n = 3$	0.9711	0.9709	0.9533	0.9537	0.9533	0.9537	0.9807	0.9806
	$n = 4$	0.8472	0.8470	0.8654	0.8653	0.8655	0.8655	0.8970	0.8969
	$n = 5$	0.7730	0.7745	0.8149	0.8152	0.8150	0.8156	0.8466	0.8476
HC	$n = 3$	0.9862	-	0.9734	-	0.9734	-	0.9908	-
	$n = 4$	0.8578	-	0.8817	-	0.8817	-	0.9046	-
	$n = 5$	0.7174	-	0.8056	-	0.8057	-	0.8089	-
DBSCAN		0.5854	-	0.4659	-	0.4241	-	0.4370	-
DGDP		0.9458	0.9399	0.9347	0.9315	0.9347	0.9316	0.9507	0.9486
S-JDP-c		0.9913	0.9883	0.9802	0.9774	0.9803	0.9774	0.9942	0.9922
S-JDP-u		0.9851	0.9843	0.9752	0.9762	0.9752	0.9762	0.9901	0.9897

Table S4: Results of C-I and AUROC from Cox PH, RSF and the proposed model.

			AUROC				
			1	5	9	13	17
Cox-LASSO	Train	0.8995	0.9220	0.9792	0.9997	0.9962	0.8765
	Test	0.8990	0.9216	0.9790	0.9997	0.9963	0.8747
RSF	Train	0.8913	0.8992	0.9745	0.9998	0.9977	0.9183
	Test	0.8870	0.9058	0.9728	0.9997	0.9970	0.8845
DGD	Train	0.8906	0.9327	0.9770	0.9998	0.9985	0.8911
	Test	0.8917	0.9300	0.9766	0.9998	0.9979	0.8854
S-JDP-c	Train	0.9186	0.9699	0.9954	0.9999	0.9993	0.9270
	Test	0.9307	0.9653	0.9958	0.9999	0.9982	0.9052
S-JDP-u	Train	0.9321	0.9699	0.9954	0.9999	0.9989	0.9042
	Test	0.9364	0.9652	0.9958	0.9999	0.9974	0.8734

Table S4: Results of AUPRC from the benchmark and the proposed models.

		AUPRC				
		1	5	9	13	17
Cox-LASSO	Train	0.0564	0.0804	0.1101	0.1272	0.3285
	Test	0.0565	0.0803	0.1099	0.1271	0.3284
RSF	Train	0.0547	0.0849	0.1150	0.1340	0.3981
	Test	0.0552	0.0848	0.1147	0.1337	0.4010
DGD	Train	0.0563	0.2303	0.3592	0.4008	0.7841
	Test	0.0562	0.2312	0.3556	0.3990	0.8055
S-JDP-c	Train	0.0536	0.2393	0.3770	0.4209	0.8108
	Test	0.0584	0.2359	0.3770	0.4227	0.8405
S-JDP-u	Train	0.0536	0.2393	0.3770	0.4209	0.8162
	Test	0.0584	0.2359	0.3770	0.4227	0.8478

S-JDP-c	Subset									
	1	2	3	4	5	6	7	8	9	10
R^2	0.8555	0.8436	0.8582	0.8577	0.8543	0.8401	0.8444	0.8539	0.8403	0.8464
MSE	0.0125	0.0142	0.0132	0.0131	0.0130	0.0144	0.0136	0.0129	0.0146	0.0146
ARI	0.9920	0.9939	0.9903	0.9880	0.9900	0.9880	0.9931	0.9887	0.9905	0.9902
AMI	0.9852	0.9886	0.9820	0.9776	0.9814	0.9776	0.9872	0.9789	0.9824	0.9817
CI	0.9673	0.9674	0.9603	0.9751	0.9652	0.9933	0.9709	0.9769	0.9644	0.9961
	Subset									
	11	12	13	14	15	16	17	18	19	20
R^2	0.8501	0.8350	0.8430	0.8520	0.8354	0.8557	0.8440	0.8287	0.8400	0.8630
MSE	0.0139	0.0143	0.0131	0.0122	0.0141	0.0123	0.0136	0.0147	0.0144	0.0126
ARI	0.9962	0.9934	0.9910	0.9942	0.9934	0.9878	0.9881	0.9884	0.9880	0.9889
AMI	0.9929	0.9877	0.9832	0.9891	0.9877	0.9774	0.9778	0.9784	0.9777	0.9794
CI	0.9637	0.9829	0.9829	0.9750	0.9529	0.9771	0.9567	0.9627	0.9606	0.9971
S-JDP-u	Subset									
	1	2	3	4	5	6	7	8	9	10
R^2	0.8616	0.8426	0.8608	0.8625	0.8573	0.8401	0.8482	0.8571	0.8418	0.8479
MSE	0.0119	0.0143	0.0130	0.0127	0.0127	0.0144	0.0133	0.0127	0.0144	0.0145
ARI	0.9838	0.9888	0.9835	0.9834	0.9838	0.9822	0.9868	0.9833	0.9836	0.9842
AMI	0.9769	0.9856	0.9761	0.9763	0.9771	0.9731	0.9821	0.9762	0.9761	0.9774
CI	0.9637	0.9850	0.9861	0.9824	0.9724	0.9979	0.9806	0.9802	0.9836	0.9968
	Subset									
	11	12	13	14	15	16	17	18	19	20
R^2	0.8531	0.8390	0.8430	0.8521	0.8384	0.8593	0.8476	0.8290	0.8479	0.8624
MSE	0.0136	0.0140	0.0131	0.0122	0.0138	0.0120	0.0133	0.0147	0.0137	0.0127
ARI	0.9886	0.9868	0.9842	0.9923	0.9828	0.9848	0.9828	0.9846	0.9862	0.9875
AMI	0.9856	0.9822	0.9777	0.9857	0.9747	0.9786	0.9749	0.9777	0.9810	0.9838
CI	0.9729	0.9939	0.9883	0.9702	0.9713	0.9812	0.9788	0.9733	0.9700	0.9892

Table S4: Survival, regression and clustering performance in each subset.

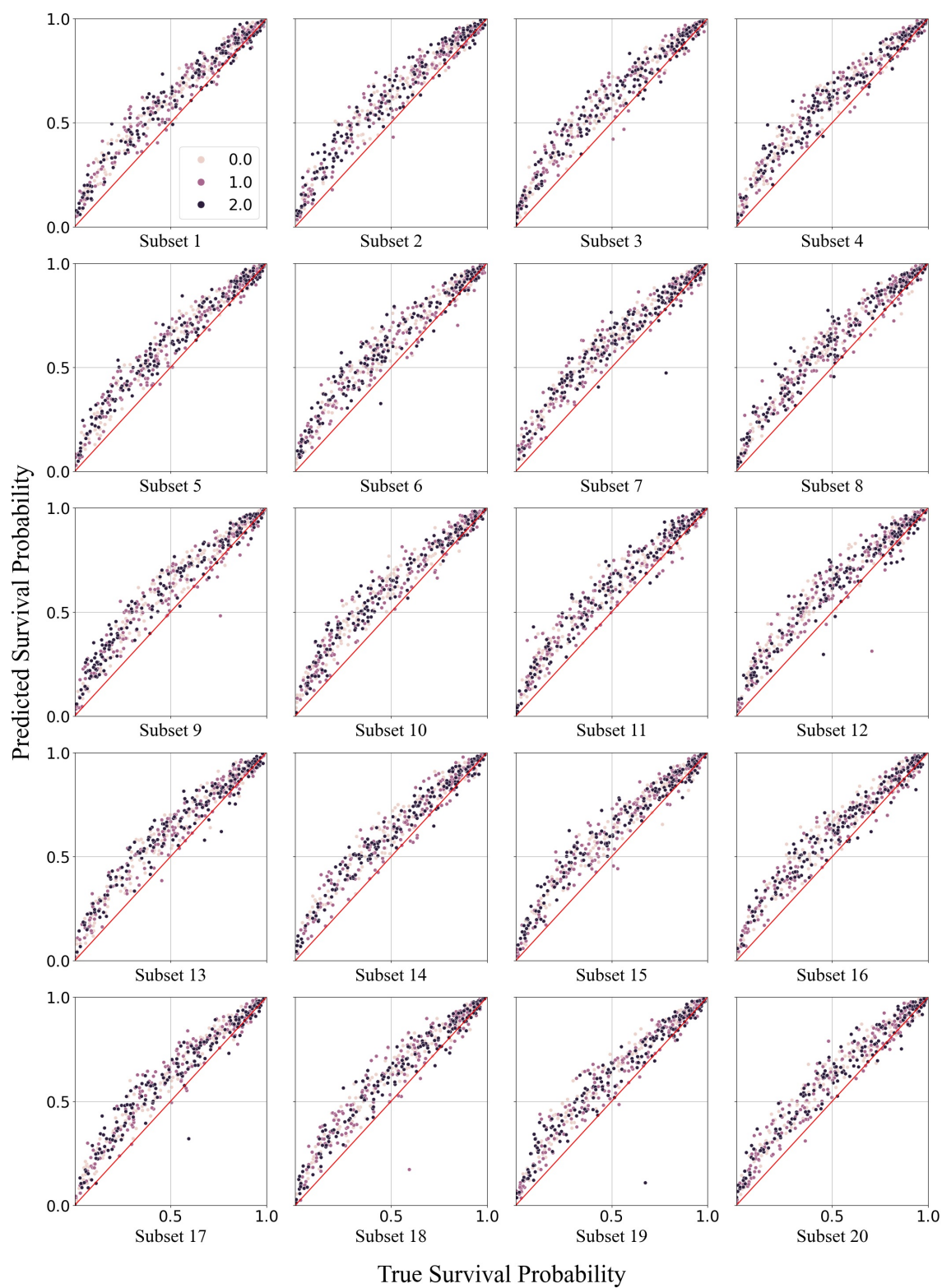
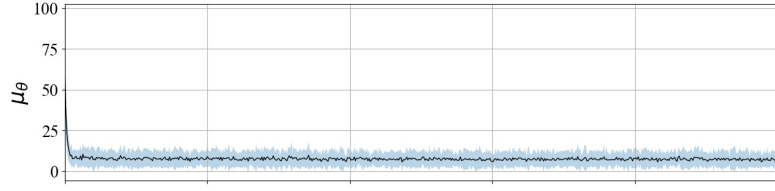
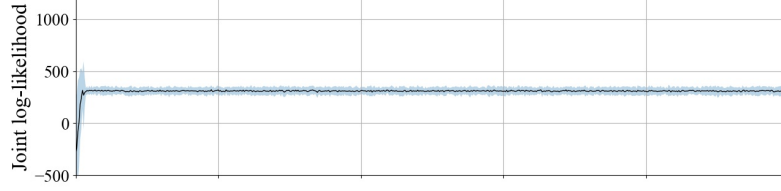


Figure S5: True Vs predicted survival probability in each subset.

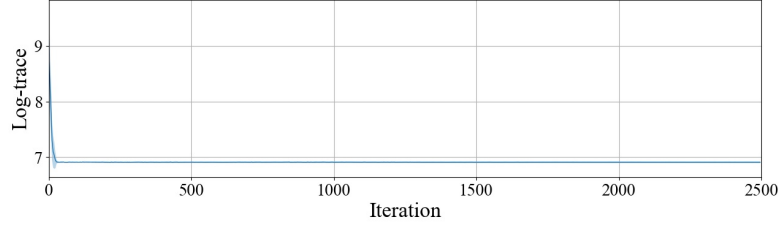
C.2 Scenario 2: $J = 5$



(a) Trend of μ_θ .

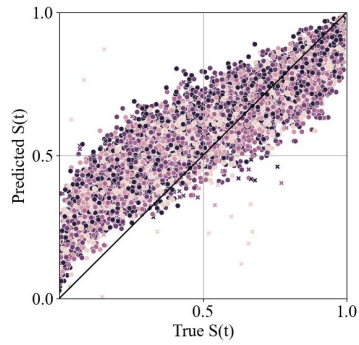


(b) Trend of joint log-likelihood.

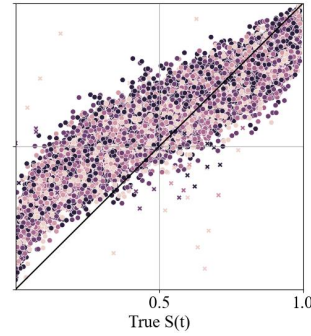


(c) Trend of log-trace of clustering allocations.

Figure S6: Convergence check.



(a) From unit-level aggregator



(b) From cluster-level aggregator

Figure S7: Predictions V.s. True survival probabilities.

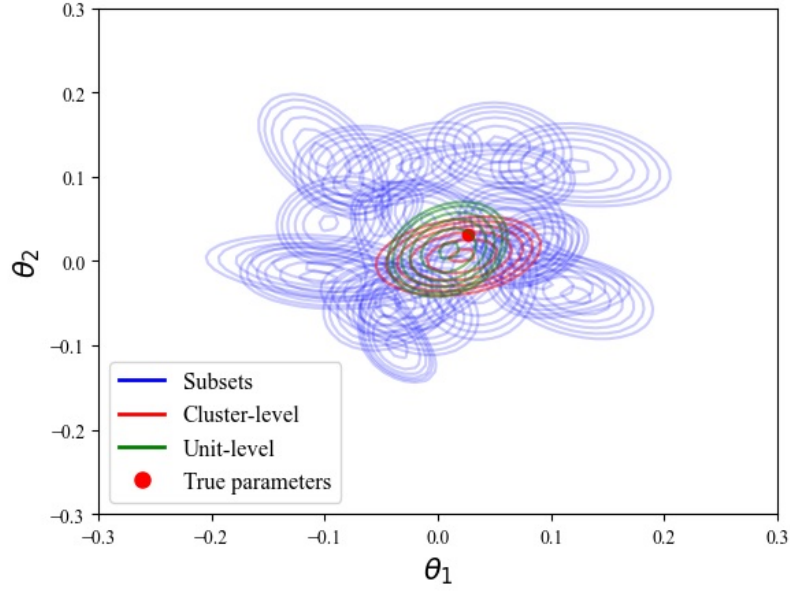


Figure S8: Kernel density estimates of subset posterior distributions and aggregated posterior distributions from cluster- and unit-level aggregators for coefficients (θ_1, θ_2) of cluster 4.

Table S8: Clustering Results from the proposed model and benchmarks.

		AR		AM		VM		FM	
		Train	Test	Train	Test	Train	Test	Train	Test
KM	$n = 4$	0.7115	0.7108	0.8180	0.8173	0.8181	0.8177	0.7909	0.7903
	$n = 5$	0.6657	0.6650	0.7915	0.7909	0.7917	0.7914	0.7481	0.7475
	$n = 6$	0.6189	0.6184	0.7654	0.7651	0.7656	0.7658	0.7066	0.7062
HC	$n = 4$	0.6714	-	0.7966	-	0.7966	-	0.7612	-
	$n = 5$	0.6262	-	0.7711	-	0.7713	-	0.7167	-
	$n = 6$	0.6214	-	0.7645	-	0.7646	-	0.7089	-
DBSCAN		0.0711	-	0.2528	-	0.2842	-	0.3494	-
DGDP		0.8756	0.8924	0.8779	0.8867	0.8781	0.8865	0.9003	0.8998
S-JDP-c		0.9134	0.9799	0.9279	0.9716	0.9279	0.9718	0.9340	0.9839
S-JDP-u		0.9800	0.9799	0.9717	0.9717	0.9718	0.9718	0.9840	0.9840

Table S8: Results of C-I and AUROC from Cox PH, RSF and the proposed model.

			AUROC				
			1	5	9	13	17
Cox-LASSO	Train	0.9327	0.9851	0.9857	0.9949	0.9952	0.9132
	Test	0.9315	0.9845	0.9854	0.9948	0.9948	0.9114
RSF	Train	0.9208	0.9806	0.9821	0.9957	0.9955	0.9295
	Test	0.9212	0.9813	0.9839	0.9953	0.9937	0.9117
DGDP	Train	0.9488	0.9865	0.9855	0.9947	0.9946	0.9489
	Test	0.9387	0.9896	0.9859	0.9943	0.9899	0.9294
S-JDP-c	Train	0.9500	0.9950	0.9909	0.9987	0.9981	0.9502
	Test	0.9563	0.9943	0.9889	0.9980	0.9968	0.9339
S-JDP-u	Train	0.9581	0.9949	0.9907	0.9988	0.9981	0.9498
	Test	0.9491	0.9943	0.9888	0.9979	0.9968	0.9330

Table S8: Results of AUPRC from the benchmark and the proposed models.

		AUPRC				
		1	5	9	13	17
Cox-LASSO	Train	0.1234	0.1515	0.1849	0.2752	0.4418
	Test	0.1236	0.1517	0.1849	0.2753	0.4420
RSF	Train	0.1250	0.1641	0.2096	0.3376	0.6302
	Test	0.1251	0.1641	0.2097	0.3375	0.6307
DGDP	Train	0.1251	0.2297	0.3255	0.5373	0.8402
	Test	0.1252	0.2311	0.3260	0.5405	0.8598
S-JDP-c	Train	0.1243	0.2404	0.3473	0.5620	0.8746
	Test	0.1309	0.2460	0.3554	0.5693	0.8970
S-JDP-u	Train	0.1243	0.2404	0.3473	0.5620	0.8747
	Test	0.1309	0.2460	0.3554	0.5693	0.8971

§-JDP-c	Subset									
	1	2	3	4	5	6	7	8	9	10
R^2	0.7586	0.7568	0.7491	0.7467	0.7593	0.7483	0.7558	0.7649	0.7279	0.7448
MSE	0.0217	0.0220	0.0222	0.0226	0.0220	0.0236	0.0209	0.0213	0.0248	0.0230
ARI	0.9413	0.9365	0.9367	0.9126	0.9149	0.9073	0.9241	0.9331	0.9253	0.9414
AMI	0.9547	0.9533	0.9531	0.9408	0.9441	0.9399	0.9472	0.9506	0.9466	0.9546
CI	0.9621	0.9481	0.9672	0.9224	0.9510	0.9596	0.9636	0.9442	0.9554	0.9599
	Subset									
	11	12	13	14	15	16	17	18	19	20
R^2	0.7478	0.7451	0.7528	0.7454	0.7423	0.7380	0.7302	0.7600	0.7657	0.7477
MSE	0.0236	0.0228	0.0218	0.0228	0.0236	0.0231	0.0244	0.0209	0.0208	0.0224
ARI	0.9457	0.9231	0.9366	0.9210	0.9322	0.9296	0.9799	0.9796	0.9798	0.9799
AMI	0.9565	0.9466	0.9532	0.9447	0.9515	0.9509	0.9720	0.9715	0.9718	0.9720
CI	0.9438	0.9531	0.9308	0.9174	0.9578	0.9435	0.9653	0.9789	0.9835	0.9822
S-JDP-u	Subset									
	1	2	3	4	5	6	7	8	9	10
R^2	0.7574	0.7588	0.7557	0.7473	0.7580	0.7505	0.7610	0.7642	0.7328	0.7464
MSE	0.0218	0.0219	0.0216	0.0226	0.0222	0.0234	0.0204	0.0214	0.0243	0.0229
ARI	0.9798	0.9798	0.9799	0.9799	0.9798	0.9799	0.9798	0.9798	0.9797	0.9796
AMI	0.9718	0.9718	0.9720	0.9720	0.9718	0.9720	0.9718	0.9718	0.9717	0.9715
CI	0.9635	0.9173	0.9639	0.9009	0.9434	0.9344	0.9465	0.9241	0.9336	0.9461
	Subset									
	11	12	13	14	15	16	17	18	19	20
R^2	0.7523	0.7430	0.7486	0.7509	0.7421	0.7350	0.7394	0.7637	0.7740	0.7512
MSE	0.0232	0.0230	0.0222	0.0223	0.0246	0.0263	0.0235	0.0206	0.0200	0.0221
ARI	0.9799	0.9796	0.9796	0.9799	0.9799	0.9800	0.9796	0.9796	0.9800	0.9799
AMI	0.9720	0.9715	0.9715	0.9720	0.9720	0.9721	0.9715	0.9715	0.9721	0.9720
CI	0.9281	0.9297	0.9178	0.9112	0.9582	0.9156	0.9525	0.9650	0.9698	0.9651

Table S8: Survival, regression and clustering performance in each subset.

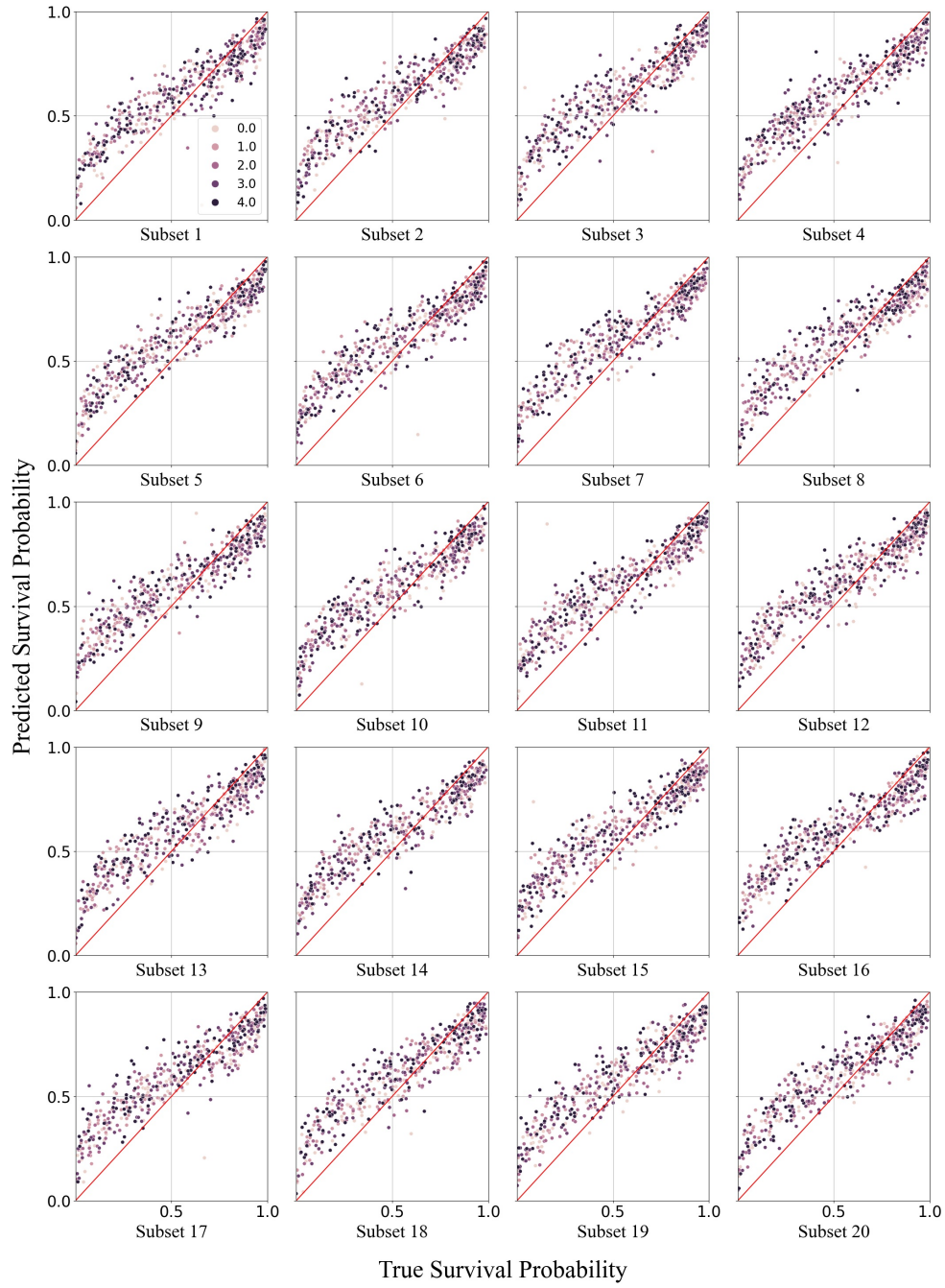
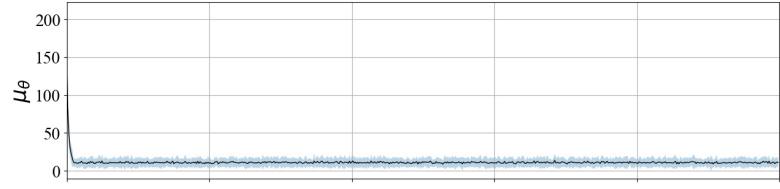
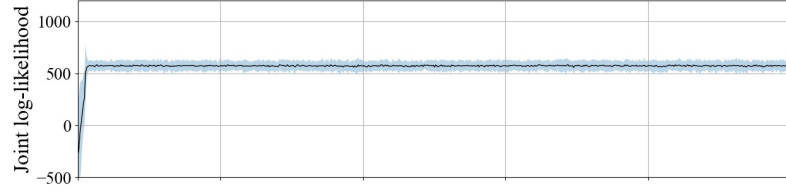


Figure S9: True Vs predicted survival probability in each subset from cluster-level aggregator.

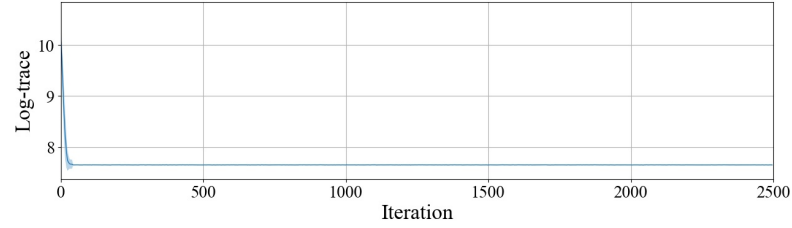
C.3 Scenario 3: $J = 7$



(a) Trend of μ_θ .

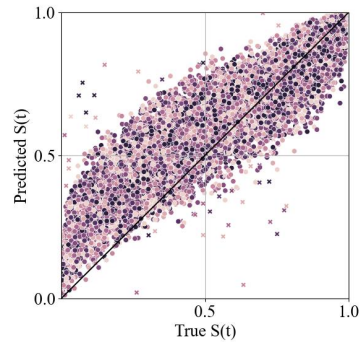


(b) Trend of joint log-likelihood.

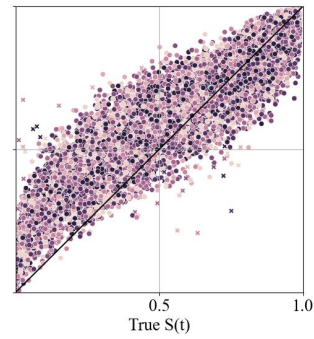


(c) Trend of log-trace of clustering allocations.

Figure S10: Convergence check.



(a) From unit-level aggregator



(b) From cluster-level aggregator

Figure S11: Predictions V.s. True survival probabilities.

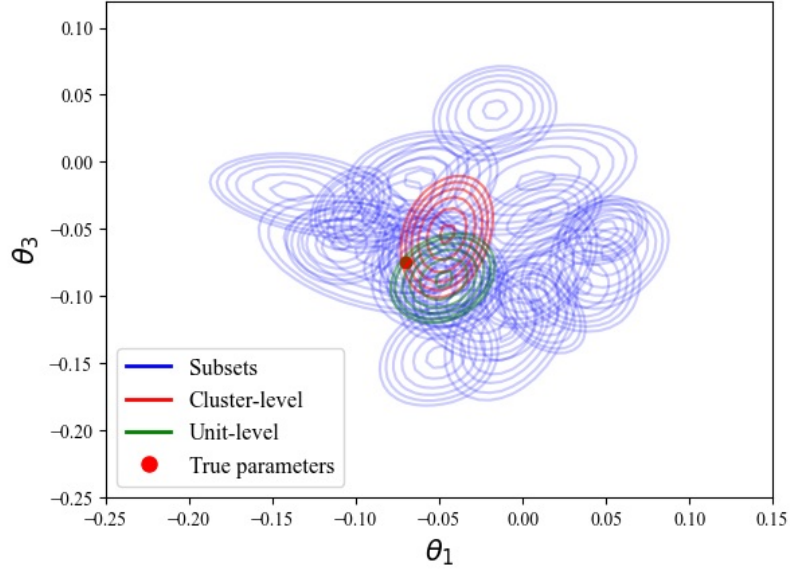


Figure S12: Kernel density estimates of subset posterior distributions and aggregated posterior distributions from cluster- and unit-level aggregators for coefficients (θ_1, θ_3) of cluster 6.

Table S12: Clustering Results from the proposed model and benchmarks.

		AR		AM		VM		FM	
		Train	Test	Train	Test	Train	Test	Train	Test
KM	$n = 6$	0.5188	0.5177	0.6464	0.6459	0.6466	0.6469	0.6033	0.6024
	$n = 7$	0.5040	0.5044	0.6391	0.6394	0.6394	0.6407	0.5795	0.5797
	$n = 8$	0.5741	0.5734	0.6661	0.6656	0.6664	0.6669	0.6344	0.6337
HC	$n = 6$	0.4989	-	0.6340	-	0.6342	-	0.5892	-
	$n = 7$	0.5881	-	0.6879	-	0.6881	-	0.6498	-
	$n = 8$	0.5957	-	0.6807	-	0.6809	-	0.6535	-
DBSCAN		0.0454	-	0.2926	-	0.3384	-	0.2384	-
DGDP		0.9517	0.9524	0.9491	0.9528	0.9492	0.9528	0.9529	0.9585
S-JDP-c		0.9740	0.9805	0.9657	0.9732	0.9657	0.9733	0.9779	0.9833
S-JDP-u		0.9806	0.9801	0.9717	0.9724	0.9717	0.9725	0.9833	0.9830

Table S12: Results of C-I and AUROC from Cox PH, RSF and the proposed model.

			AUROC				
			1	5	9	13	17
Cox-LASSO	Train	0.9291	0.9618	0.9925	0.9798	0.9781	0.9388
	Test	0.9285	0.9614	0.9923	0.9795	0.9771	0.9367
RSF	Train	0.9234	0.9119	0.9884	0.9834	0.9780	0.9193
	Test	0.9210	0.9110	0.9888	0.9817	0.9753	0.9191
DGDP	Train	0.9504	0.9769	0.9887	0.9798	0.9756	0.9515
	Test	0.9492	0.9761	0.9882	0.9779	0.9727	0.9501
S-JDP-c	Train	0.9684	0.9799	0.9985	0.9968	0.9952	0.9839
	Test	0.9698	0.9755	0.9981	0.9956	0.9939	0.9870
S-JDP-u	Train	0.9634	0.9798	0.9984	0.9968	0.9951	0.9839
	Test	0.9544	0.9757	0.9981	0.9956	0.9939	0.9870

Table S12: Results of AUPRC from the benchmark and the proposed models.

		AUPRC				
		1	5	9	13	17
Cox-LASSO	Train	0.0255	0.0348	0.0604	0.1405	0.3181
	Test	0.0255	0.0348	0.0604	0.1405	0.3184
RSF	Train	0.0251	0.0352	0.0619	0.1493	0.3430
	Test	0.0251	0.0352	0.0619	0.1495	0.3474
DGDP	Train	0.0255	0.1494	0.3310	0.6173	0.8264
	Test	0.0250	0.1502	0.3297	0.6295	0.8289
S-JDP-c	Train	0.0255	0.1571	0.3572	0.6408	0.8715
	Test	0.0236	0.1563	0.3556	0.6522	0.8743
S-JDP-u	Train	0.0255	0.1571	0.3572	0.6408	0.8715
	Test	0.0236	0.1563	0.3552	0.6521	0.8743

S-JDP-c	Subset									
	1	2	3	4	5	6	7	8	9	10
R^2	0.7921	0.8027	0.7895	0.8008	0.7887	0.8058	0.7846	0.7685	0.7871	0.8069
MSE	0.0190	0.0168	0.0183	0.0182	0.0195	0.0171	0.0191	0.0204	0.0189	0.0173
ARI	0.9761	0.9779	0.9748	0.9747	0.9726	0.9749	0.9750	0.9737	0.9786	0.9748
AMI	0.9752	0.9750	0.9737	0.9735	0.9734	0.9739	0.9740	0.9751	0.9785	0.9737
CI	0.9662	0.9768	0.9500	0.9888	0.9620	0.9645	0.9737	0.9842	0.9545	0.9710
	Subset									
	11	12	13	14	15	16	17	18	19	20
R^2	0.8114	0.7935	0.7959	0.7954	0.7701	0.7763	0.7980	0.7857	0.7633	0.7895
MSE	0.0168	0.0191	0.0185	0.0177	0.0195	0.0187	0.0191	0.0198	0.0220	0.0190
ARI	0.9777	0.9755	0.9753	0.9814	0.9740	0.9774	0.9802	0.9801	0.9807	0.9804
AMI	0.9748	0.9745	0.9744	0.9768	0.9727	0.9745	0.9754	0.9753	0.9760	0.9756
CI	0.9735	0.9718	0.9565	0.9795	0.9784	0.9673	0.9697	0.9579	0.9316	0.9621
S-JDP-u	Subset									
	1	2	3	4	5	6	7	8	9	10
R^2	0.7950	0.8045	0.7886	0.8033	0.7890	0.8047	0.7915	0.7676	0.7865	0.8141
MSE	0.0231	0.0216	0.0233	0.0215	0.0245	0.0193	0.0228	0.0248	0.0225	0.0202
ARI	0.9812	0.9801	0.9806	0.9803	0.9807	0.9806	0.9810	0.9808	0.9799	0.9806
AMI	0.9766	0.9753	0.9759	0.9755	0.9759	0.9758	0.9763	0.9761	0.9750	0.9759
CI	0.9708	0.9750	0.9465	0.9869	0.9553	0.9589	0.9640	0.9764	0.9517	0.9682
	Subset									
	11	12	13	14	15	16	17	18	19	20
R^2	0.8158	0.7949	0.7934	0.8029	0.7772	0.7804	0.8028	0.7958	0.7795	0.7897
MSE	0.0199	0.0225	0.0223	0.0213	0.0210	0.0205	0.0186	0.0189	0.0205	0.0190
ARI	0.9801	0.9801	0.9800	0.9807	0.9800	0.9801	0.9802	0.9797	0.9801	0.9802
AMI	0.9753	0.9752	0.9751	0.9760	0.9752	0.9753	0.9754	0.9747	0.9752	0.9753
CI	0.9709	0.9670	0.9531	0.9776	0.9688	0.9586	0.9795	0.9761	0.9551	0.9848

Table S12: Survival, regression and clustering performance in each subset.

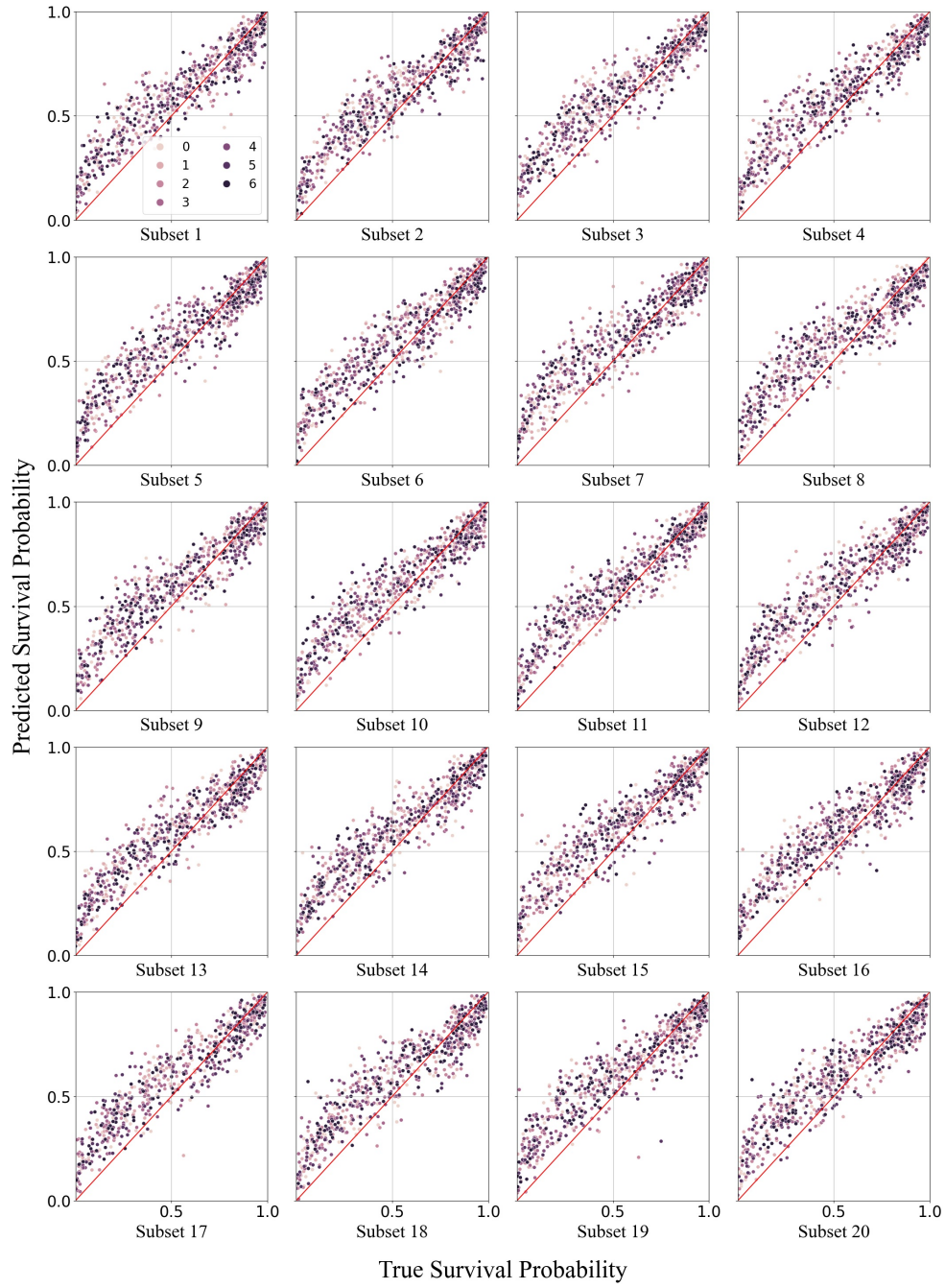
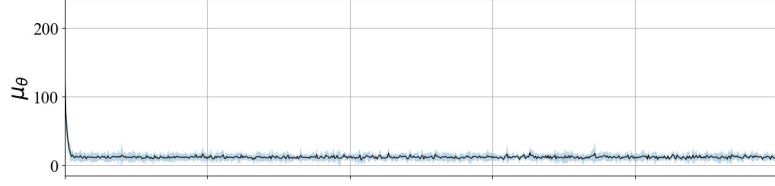
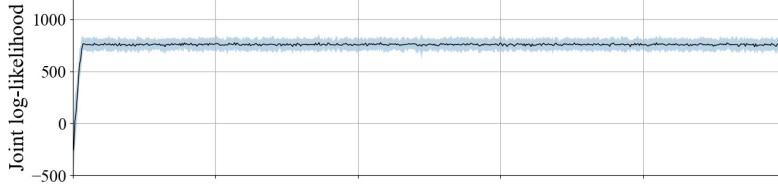


Figure S13: True Vs predicted survival probability in each subset from cluster-level aggregator.

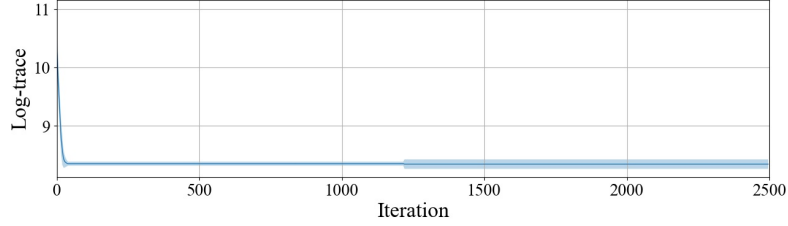
C.4 Scenario 4: $J = 9$



(a) Trend of μ_θ .

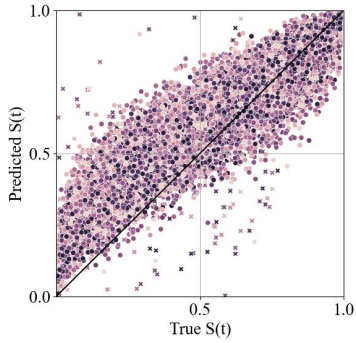


(b) Trend of joint log-likelihood.

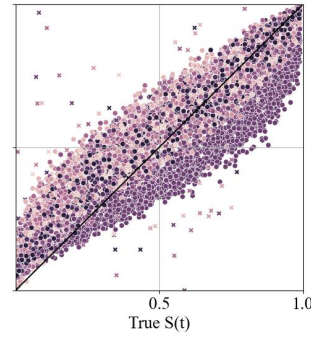


(c) Trend of log-trace of clustering allocations.

Figure S14: Convergence check.



(a) From unit-level aggregator



(b) From cluster-level aggregator

Figure S15: Predictions V.s. True survival probabilities.

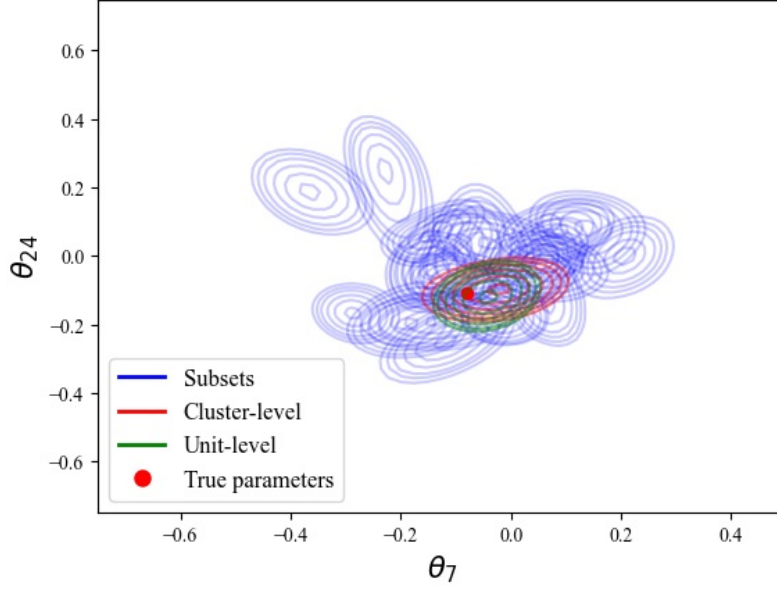


Figure S16: Kernel density estimates of subset posterior distributions and aggregated posterior distributions from cluster- and unit-level aggregators for coefficients (θ_7, θ_{24}) of cluster 4.

Table S16: Clustering Results from the proposed model and benchmarks.

		AR		AM		VM		FM	
		Train	Test	Train	Test	Train	Test	Train	Test
KM	$n = 8$	0.4902	0.4898	0.7121	0.7120	0.7123	0.7131	0.5870	0.5866
	$n = 9$	0.5909	0.5902	0.7553	0.7548	0.7555	0.7558	0.6517	0.6510
	$n = 10$	0.5929	0.5927	0.7479	0.7475	0.7482	0.7487	0.6414	0.6411
HC	$n = 8$	0.5027	-	0.7156	-	0.7158	-	0.5993	-
	$n = 9$	0.6419	-	0.7762	-	0.7764	-	0.6945	-
	$n = 10$	0.6184	-	0.7543	-	0.7545	-	0.6662	-
DBSCAN		0.1360	-	0.3748	-	0.4169	-	0.3061	-
DGDP		0.9452	0.9477	0.9406	0.9476	0.9403	0.9475	0.9459	0.9500
S-JDP-c		0.9730	0.9813	0.9687	0.9774	0.9688	0.9775	0.9761	0.9833
S-JDP-u		0.9819	0.9819	0.9770	0.9775	0.9770	0.9776	0.9840	0.9839

Table S16: Results of C-I and AUROC from Cox PH, RSF and the proposed model.

C-I			AUROC				
			1	5	9	13	17
Cox-LASSO	Train	0.9211	0.9644	0.9788	0.9963	0.9951	0.8676
	Test	0.9200	0.9631	0.9785	0.9959	0.9949	0.8669
RSF	Train	0.9268	0.9643	0.9834	0.9997	0.9972	0.9060
	Test	0.9239	0.9642	0.9814	0.9997	0.9965	0.8821
DGBP	Train	0.9207	0.9768	0.9905	0.9918	0.9961	0.9293
	Test	0.9225	0.9699	0.9850	0.9879	0.9854	0.8999
S-JDP-c	Train	0.9308	0.9883	0.9910	0.9973	0.9957	0.9424
	Test	0.9337	0.9823	0.9849	0.9907	0.9854	0.9102
S-JDP-u	Train	0.9075	0.9905	0.9953	0.9999	0.9995	0.9457
	Test	0.9104	0.9884	0.9950	0.9999	0.9990	0.9232

Table S16: Results of AUPRC from the benchmark and the proposed models.

		AUPRC				
		1	5	9	13	17
Cox-LASSO	Train	0.0849	0.1311	0.1600	0.2146	0.4185
	Test	0.0850	0.1313	0.1602	0.2147	0.4187
RSF	Train	0.0848	0.1374	0.1788	0.2535	0.6359
	Test	0.0848	0.1374	0.1788	0.2536	0.6372
DGBP	Train	0.0844	0.2618	0.3672	0.5010	0.8274
	Test	0.0845	0.2649	0.3740	0.5055	0.8295
S-JDP-c	Train	0.0842	0.2842	0.3938	0.5301	0.8769
	Test	0.0849	0.2873	0.3978	0.5361	0.8802
S-JDP-u	Train	0.0840	0.2836	0.3930	0.5290	0.8765
	Test	0.0842	0.2850	0.3944	0.5309	0.8782

S-JDP-c	Subset									
	1	2	3	4	5	6	7	8	9	10
R^2	0.8524	0.8409	0.7488	0.7701	0.7809	0.7919	0.8383	0.7929	0.8264	0.8215
MSE	0.0137	0.0147	0.0220	0.0199	0.0195	0.0196	0.0142	0.0197	0.0149	0.0162
ARI	0.9756	0.9775	0.9735	0.9756	0.9711	0.9756	0.9780	0.9754	0.9734	0.9760
AMI	0.9758	0.9762	0.9752	0.9756	0.9742	0.9758	0.9765	0.9755	0.9749	0.9758
CI	0.9482	0.9570	0.9375	0.9449	0.9123	0.9438	0.9572	0.9280	0.9379	0.9365
	Subset									
	11	12	13	14	15	16	17	18	19	20
R^2	0.7961	0.8014	0.7952	0.7644	0.8333	0.8353	0.8384	0.8448	0.8388	0.8392
MSE	0.0184	0.0181	0.0186	0.0214	0.0158	0.0144	0.0140	0.0141	0.0145	0.0150
ARI	0.9761	0.9750	0.9753	0.9738	0.9729	0.9760	0.9812	0.9810	0.9811	0.9808
AMI	0.9760	0.9754	0.9756	0.9752	0.9747	0.9758	0.9783	0.9782	0.9783	0.9779
CI	0.8718	0.9649	0.8962	0.9163	0.9668	0.9357	0.9260	0.8970	0.8603	0.9164
S-JDP-u	Subset									
	1	2	3	4	5	6	7	8	9	10
R^2	0.8144	0.8029	0.7765	0.7855	0.8064	0.7994	0.7894	0.7847	0.8019	0.8037
MSE	0.0221	0.0232	0.0246	0.0236	0.0223	0.0239	0.0236	0.0255	0.0220	0.0228
ARI	0.9818	0.9809	0.9820	0.9821	0.9814	0.9820	0.9817	0.9820	0.9820	0.9822
AMI	0.9786	0.9781	0.9788	0.9789	0.9784	0.9786	0.9786	0.9790	0.9789	0.9790
CI	0.9391	0.9273	0.9264	0.9393	0.8942	0.9346	0.9351	0.9240	0.9254	0.9350
	Subset									
	11	12	13	14	15	16	17	18	19	20
R^2	0.8057	0.7740	0.8004	0.7901	0.8137	0.7895	0.7903	0.8059	0.7970	0.7974
MSE	0.0225	0.0256	0.0231	0.0240	0.0227	0.0234	0.0182	0.0176	0.0183	0.0190
ARI	0.9816	0.9818	0.9819	0.9821	0.9811	0.9817	0.9817	0.9809	0.9820	0.9822
AMI	0.9784	0.9786	0.9788	0.9787	0.9782	0.9785	0.9786	0.9781	0.9788	0.9790
CI	0.8566	0.9392	0.8805	0.9081	0.9558	0.9269	0.9247	0.9284	0.9128	0.9180

Table S16: Survival, regression and clustering performance in each subset.

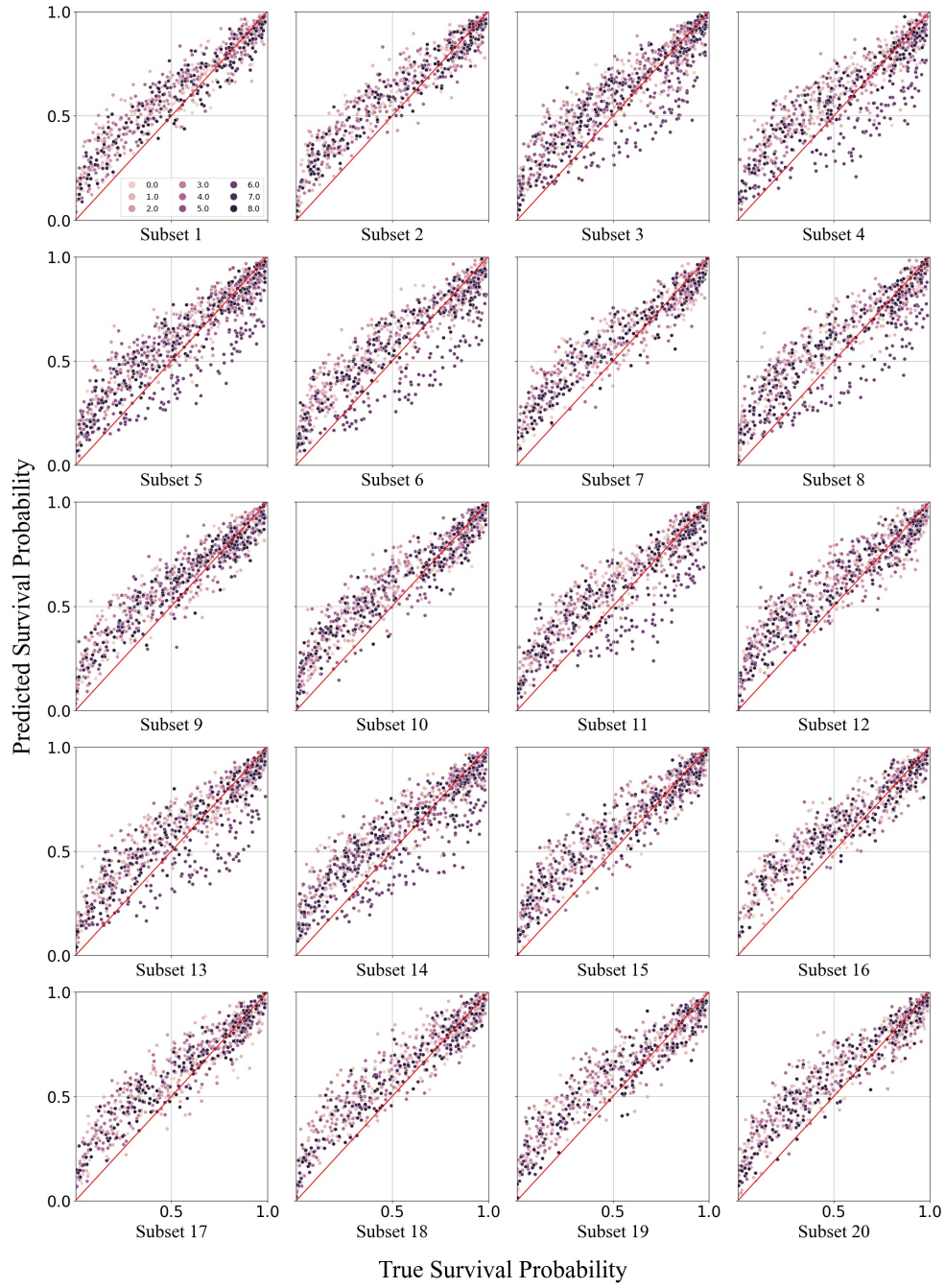


Figure S17: True Vs predicted survival probability in each subset from cluster-level aggregator.

D Conditional V.s. Marginal Methods

To compare the performance of conditional and marginal methods, we pick one subset in each scenario and use the deviance D [1] to measure the algorithmic performance:

$$D = -2 \sum_{i=1}^n \log \left\{ \sum_{j=1}^{\max s_y} \frac{n_j}{n} k(y_i | \mathbf{x}_i, \boldsymbol{\theta}_j) \right\}.$$

As shown in Figure S18, in our settings, the marginal method slightly outperforms the conditional one which is consistent with the conclusion in [2].

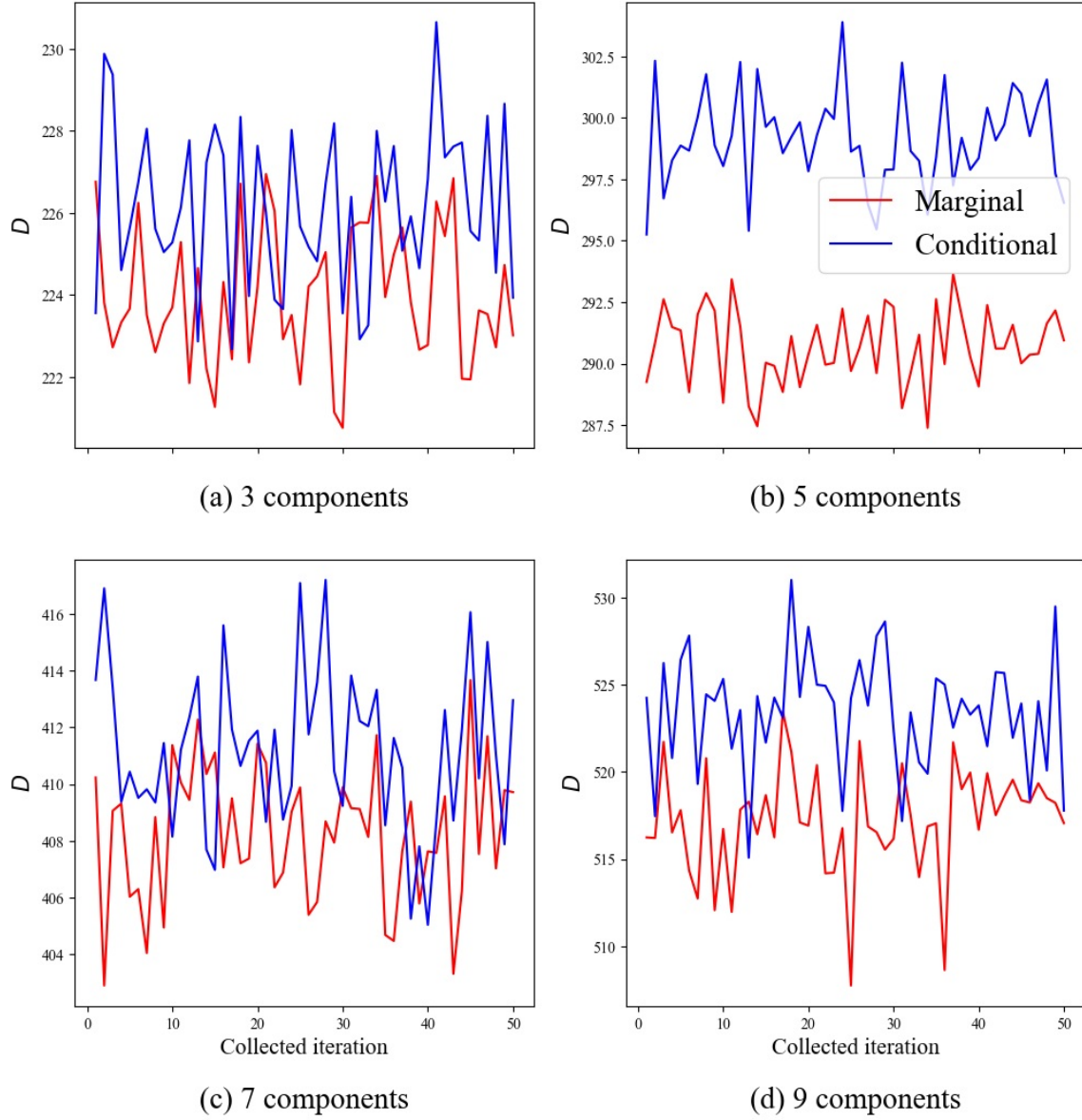


Figure S18: Deviance of conditional and marginal methods over 50 collected iterations.

E Case Study

Figure S19 shows the trends of the interested variables over 3000 iterations and demonstrates the algorithm convergence before collecting the subset posterior samples.

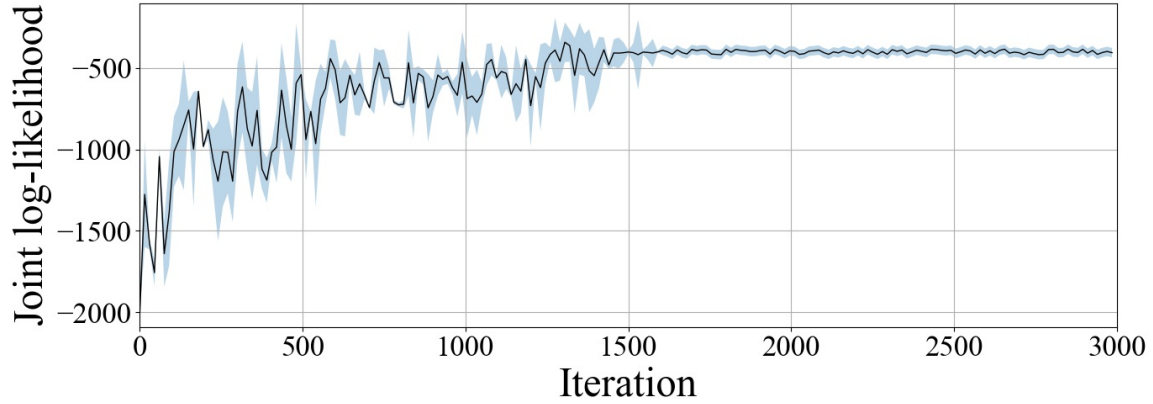


Figure S19: Convergence check for case study.

References

- [1] P. J. Green and S. Richardson, “Modelling heterogeneity with and without the dirichlet process,” *Scandinavian journal of statistics*, vol. 28, no. 2, pp. 355–375, 2001.
- [2] O. Papaspiliopoulos and G. O. Roberts, “Retrospective markov chain monte carlo methods for dirichlet process hierarchical models,” *Biometrika*, pp. 169–186, 2008.